

Trust in the digital age: Chatbot service quality and its influence on e-WOM and loyalty among consumers

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ABSTRACT

The quality of information and systems about chatbots is a relevant issue among consumers because of its potential contribution to trust and satisfaction and, finally, loyalty and electronic Word-of-Mouth (e-WOM). Understanding the factors that have an effect on the variables can be used to create new services. This study, based on Uncertainty Reduction Theory, Expectation-Disconfirmation Theory, Self-Determination Theory, and Attribution Theory, analyzed the constructs to understand consumers' perceptions. The research is a non-experimental and correlational study involving 362 bank consumers in Peru. A multivariate technique (PLS-SEM) was employed to test the proposed model. Information quality and system quality had a positive and significant effect on trust. Trust, in turn, had an effect on satisfaction, and finally, satisfaction showed a positive and significant effect on loyalty and e-WOM. The findings of this study provide valuable recommendations for enhancing the customer experience through chatbots across various industries. It highlights the need to invest in technologies that enhance information quality and chatbot performance, as these elements increase user satisfaction and trust. Companies can enhance the perception of quality through intuitive interface design and adaptive responses that effectively resolve user concerns, promoting a trusted and comfortable interaction environment. Another key implication is the relevance of chatbots for building loyalty and generating positive e-WOM.

1. Introduction

In the digital age, the interaction between brands and consumers has undergone a profound transformation, especially through the use of artificial intelligence technologies, such as chatbots. These systems, which enable rapid, automated communication, have become fundamental to enhancing the customer experience by providing real-time services and personalized responses. However, the success of these interactions largely depends on the perceived quality of the service chatbots provide, as this directly influences the trust consumers place in the technology and, by extension, in the brand it represents. Chatbots are intelligent systems that facilitate interaction between computers and people, and they have become widely used nowadays, especially in banking systems [1]. Humans are becoming less willing to visit bank branches to carry out the various procedures they require, as doing so involves time and effort. This is why banks have begun offering digital tools, such as chatbots [2].

The use of chatbots significantly enhances the user experience, making it necessary to strengthen their reliability and responsiveness to

improve the customer experience and gain a competitive advantage over non-users [3]. Chatbots have become a new channel of contact between customers and banks by enabling automatic learning to the point of using natural language, adapted and personalized to each user, thereby improving their performance. However, there are still major challenges, such as ethical considerations, biases, and limitations inherent to any technology. Therefore, the new human-computer interaction through chatbots is beginning to reveal a changing landscape [4]. Chatbots, also known as conversational agents, are becoming increasingly relevant across sectors, automating how they communicate with customers. It is estimated that the market size will reach \$1.23 billion worldwide by the end of 2025, resulting in increased interaction among people [5].

Trust, in this context, is a crucial factor in fostering loyalty and e-WOM behavior, which in turn can strengthen a brand's position in an increasingly competitive marketplace. When consumers perceive high quality in their interactions with chatbots, they are more likely to trust the company and recommend it through digital channels, thereby contributing to a positive brand image. However, there is a lack of consensus on the different effects that shape satisfaction and the

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variables derived from it. For this reason, the present study will investigate how information quality and system quality have an effect on satisfaction and, in turn, how this has an effect on consumer trust, which generates loyalty and e-WOM.

The literature consistently establishes that information and system quality in digital services are antecedents of user trust and satisfaction, which in turn drive loyalty and e-WOM in various contexts, including banking chatbots [6–8]; however, two significant gaps remain, as there is a fragmented understanding of the integrated psychological and theoretical mechanisms that explain how and why these relationships develop specifically in chatbot interactions. Current research lacks a theoretical synthesis that holistically explains the process, from technical quality to relational outcomes. Similarly, empirical findings show inconsistencies, such as non-significant effects of quality on trust, suggesting the need to examine these mechanisms in different contexts. This study addresses these gaps by proposing and testing a comprehensive model grounded in four complementary theories, Uncertainty Reduction Theory, Expectation Disconfirmation Theory, Self-Determination Theory, and Attribution Theory, that explain the underlying socio-cognitive and motivational mechanisms. We apply this model to the context of banking chatbots in Peru, contributing evidence from an emerging economy and testing a serial mediation model that clarifies the sequence quality trust satisfaction loyalty/e-WOM.

2. Related studies on service quality in chatbots, trust, and outcomes

Previous research has established consistent relationships between the constructs of our model. Regarding trust, multiple studies confirm that information and system quality are significant predictors. For example, Nelloh et al. [9] and Setiyono et al. [10] found positive effects of information quality on trust in financial applications, while Ludwina et al. [11] and Chandra and Tan [12] validated similar effects for system quality. With respect to trust and satisfaction, there is a broad consensus on their positive relationship, as demonstrated across diverse banking contexts [13,14]. The literature consistently supports that satisfaction leads to loyalty [15,16] and generates positive e-WOM [17,18]. With respect to trust and satisfaction, there is a broad consensus on their positive relationship, as demonstrated across diverse banking contexts [13,14]. The literature consistently supports that satisfaction leads to loyalty [15,16] and generates positive e-WOM [17,18].

Despite this general consensus, some findings present inconsistencies that require further exploration. In particular, the relationship between quality and trust shows variations. Manan et al. [19] found no significant effect of system quality, and Chandra and Tan [12] reported a non-significant effect of information quality in teleconsultations. These discrepancies suggest that contextual, cultural, or mediating factors could moderate these relationships. Given this landscape, the present research seeks to provide clarity by integrating a multifaceted theoretical framework that explains the mechanisms underlying these relationships, particularly useful for understanding the observed inconsistencies; the examination of serial mediation mechanisms (trust and satisfaction) that explain how quality is transmitted to relational outcomes, a less explored aspect in chatbots; and the application of this integrated model to the context of banking chatbots in an emerging economy, where the literature is still incipient.

3. Theoretical framework and hypotheses

3.1. Theories

This study is based on the complementary integration of four theories that together explain the psychological and cognitive processes through which a chatbot's technical quality influences relational outcomes. Each theory illuminates a distinct stage or mechanism within the proposed sequence of effects, thereby justifying the hypotheses.

3.1.1. Uncertainty Reduction Theory (URT)

This theory provides the foundation for the initial stage of trust formation in human-technology interactions, where uncertainty is inherent, as chatbots can introduce ambiguity due to their automated nature. URT posits that users seek clear signals to predict a conversational agent's behavior [20]. In the proposed research model, high-quality information (clarity, accuracy) and system quality (reliability, responsiveness) serve as mechanisms to reduce uncertainty, mitigating user anxiety and paving the way for the development of initial trust.

3.1.2. Attribution Theory (AT)

Attribution Theory, first proposed by Heider [21] and extended by Weiner [22], explains how people interpret the causes of events and behaviors, and how these attributions affect their emotions and future decisions. AT also complements URT by explaining how users causally interpret chatbot performance. If the interaction is successful, the user must attribute this outcome to both internal factors, such as the chatbot's technological competence and commitment to quality, and external factors, such as luck or temporary conditions. This theory is particularly relevant for chatbots, where perceived quality can be directly attributed to the organization behind the technology.

3.1.3. Self-Determination Theory (SDT)

SDT is a theoretical framework developed by Deci and Ryan [23] that explains human motivation in terms of the satisfaction of three fundamental psychological needs: autonomy, competence, and relatedness. An effective chatbot can satisfy the need for competence by enabling users to resolve queries; likewise, it can satisfy the need for autonomy by offering control over the interaction. Satisfying these intrinsic needs increases overall satisfaction and can reinforce trust by fostering a sense of empowerment and personal efficacy in using technology.

3.1.4. Expectation disconfirmation theory (EDT)

EDT is a theory that explains how consumers' perceptions of a product or service affect their satisfaction [24,25]. EDT allows for structuring the final cognitive evaluation that leads to loyalty and e-WOM. Likewise, trust, formed via URT and AT, and the satisfaction of needs, via SDT, raise user expectations. Based on this, EDT posits that when the chatbot's actual performance meets or exceeds these high expectations, it results in robust satisfaction. This satisfaction, in turn, is the direct antecedent of the behavioral intention to reuse the service and to recommend it, thus completing the cycle from technical quality to business results.

3.2. Variables

3.2.1. Information quality

Quality is defined as an indicator of how well a previously established task has been achieved. It is also defined as the ability of a service or product to achieve consumer satisfaction [26]. The quality of information is the measure or degree of information that is produced with certain characteristics such as efficiency, easy access, timely, reliable, accurate, flexible, data quality, sophisticated, portable, complete, and customizable, among others, in order to be valuable for a customer or user to make a better decision [27].

Information quality enables the implementation and adherence to software standards, thereby achieving best practices, enhanced performance, and error avoidance. The value of information quality lies in providing useful, user-friendly applications for informed decision-making. Therefore, the quality of information is determined by implementing systems that adapt to users and conform to the industry or environmental standards in which they operate. The idea is not only to meet a customer's expectations but also to exceed them in every interaction, thereby improving results, understanding, and decision-making [28]. Properly managing information quality positively impacts

organizational performance [29].

3.2.2. System quality

The quality of the system has a notable impact on the satisfaction of companies and users; people need to be sure that the system's quality aligns with what they really need and want. The system must be continuously improved, and for this to happen, user feedback plays a fundamental role. The quality of the system can be measured. Therefore, it can always be improved; having quality controls for a system ex-ante, during, and after should be essential [30]. System quality management involves a comprehensive strategy in which each stage of the system's life cycle is evaluated and optimized. For a system to be truly effective and meet user and business expectations, it is essential to implement a continuous improvement approach grounded in both quantitative and qualitative indicators. These indicators allow for the identification of areas of opportunity and strengths, enabling the system's functionality to be adapted to the changing needs and desires of users and market requirements [31].

3.2.3. Consumer satisfaction

Consumer satisfaction is a critical component in the digital age, especially in the service sector. In a world where technology is ubiquitous, consumers have higher expectations than ever before regarding service quality and overall experience [32]. Customer satisfaction is the evaluation of the shopping experience relative to prior expectations. In the digital environment, these expectations can be influenced by multiple factors, including service quality, platform ease of use, and customer support availability [33]. One of the key aspects affecting consumer satisfaction in the digital environment is service quality. The perceived quality of an online service is the consumer's evaluation of its effectiveness and efficiency. In the digital environment, where interactions are often asynchronous and mediated by technology, service quality becomes a crucial factor. Thus, chatbots, which are increasingly used to interact with consumers, must provide accurate and fast responses to meet user expectations. The quality of service provided by these tools can significantly influence satisfaction [30].

3.2.4. Trust

Consumer trust is the consumer's belief that a seller or service provider will deliver quality products and fulfill promises [34]. This trust is built over time through past experiences, brand reputation, third-party recommendations and marketing communication. Trust refers to the supplier's integrity, competence and ability to meet consumer expectations [35]. In situations of uncertainty, consumers tend to minimize perceived risks by opting for brands they trust. Trust acts as a buffer, allowing consumers to feel secure when transacting [36]. Consumer trust affects purchasing decisions, willingness to try new products, and willingness to pay premium prices; consumers who trust a brand are also more likely to consider its products when making purchasing decisions. In a world where offers are everywhere, a key differentiator is the consumer's personal experience with a brand or product. If this is achieved, it has a positive impact on the consumer, developing a trust that can be medium or long-term as long as it continues to be reinforced by good interactions with the brand [37].

3.2.5. Consumer loyalty

For a consumer to feel loyal to a brand, they need to trust it. Trust is the foundation on which all meaningful relationships are built; consumers want to know that they will not be disappointed [38]. This trust process involves a perception of authenticity and credibility. If the brand fails, the chances of redemption are minimal; thus, trust is non-negotiable, as it is what keeps loyalty in place. Consumer loyalty extends beyond a simple preference for a brand; it is a complex mix of emotions, memories, and rational decisions [39]. For consumers, choosing a brand means taking on certain values that reflect their own identity and personality; when a consumer chooses a product or service

at one company over another company, they are making a choice and, at the same time, paying for a brand experience that allows them to express themselves as an individual or as part of a community.

3.2.6. Electronic word of mouth (e-WOM)

e-WOM primarily refers to the exchange of information and/or knowledge among users or consumers after purchasing and using an electronic product [40]. Consumers share their personal experiences with a particular product, which can influence other users. Social networks have enabled more people to express their opinions about products and services. Marketers are focusing on maximizing the value of e-WOM because it provides insight into what the market thinks of the company and helps [41]. If it is addressed promptly, it can improve the company's performance in a targeted way.

3.3. Effects between variables

3.3.1. Effect of information and system quality on trust

Based on URT, we argue that high-quality information reduces perceived ambiguity, while high-quality systems make interactions more predictable. Together, they minimize initial uncertainty, creating conditions conducive to trust [20]. Complementarily, from Attribution Theory, consistent performance leads users to attribute positive and trustworthy capabilities to the company [22]. This is particularly relevant in chatbots, where system quality is directly observable [8].

H1: Information quality has a positive and significant effect on user trust in the chatbot.

H2: System quality has a positive and significant effect on user trust in the chatbot

3.3.2. Effect of trust on satisfaction

Trust establishes positive expectations about future interactions. According to EDT, when these high expectations are confirmed by experience, satisfaction results [25]. Simultaneously, according to SDT, trust reduces anxiety and allows the user to focus on satisfying their needs for competence and autonomy during the interaction, thus enhancing intrinsic satisfaction [23].

H3: Trust has a positive and significant effect on user satisfaction with the chatbot.

3.3.3. Effect of satisfaction on loyalty and e-WOM

According to EDT, satisfaction resulting from confirmed or exceeded expectations is the direct antecedent of positive behavioral intentions [25]. A satisfied user will tend to reuse the service and share their positive experience as a form of social validation and reciprocity. AT reinforces that if users attribute their satisfaction to the company's stable characteristics, they are more likely to engage with and recommend it.

H4: Satisfaction has a positive and significant effect on loyalty to the chatbot service.

H5: Satisfaction has a positive and significant effect on the generation of positive e-WOM.

3.3.4. Serial mediation

Integrating all the theories, we propose that trust and satisfaction act as sequential mechanisms that transmit the effects of quality to outcomes. Trust and satisfaction serially mediate the relationships between information quality and system quality with loyalty and e-WOM. This analysis establishes the following hypotheses, which are tested using total indirect effects.

H6: Trust mediates the relationship between information quality and satisfaction

H7: Trust mediates the relationship between system quality and satisfaction

H8: Trust and satisfaction mediate the relationship between information quality and loyalty

H9: Trust and satisfaction mediate the relationship between information quality and e-WOM

H10: Trust and satisfaction mediate the relationship between system quality and loyalty

H11: Trust and satisfaction mediate the relationship between system quality and e-WOM

H12: Satisfaction mediates the relationship between trust and loyalty

H13: Satisfaction mediates the relationship between trust and e-WOM

4. Methodology

4.1. Design

This study uses a quantitative approach with a correlational design, focusing on analysis of numerical data that enable an in-depth understanding of the phenomena under study. The quantitative approach enables the examination of patterns and trends through statistical analysis. By opting for a correlational design, significant associations between these variables can be identified without the need for experimental intervention. This is ideal for investigating how the quality of systems and information affects user trust and satisfaction, as well as their influence on e-WOM and loyalty.

4.2. Measurement scale

The questionnaire was used to collect data and consisted of 26 items on a 5-point Likert scale, where 1 means "Totally disagree" and 5 means "Totally agree." The number of items for the variables is described next: Six items for information quality, four from Trivedi [42] and two from Zhang et al. [43]; five items for system quality, four from Trivedi [42] and one proposed by the authors; four items for satisfaction [44], four for trust Rajaobelina et al. [45], four for loyalty (proposed by authors, and three for e-WOM [46]. The total number of items is described in [Appendix 1](#).

4.3. Data collection

Each participant was required to provide informed consent before beginning, during which they were presented with a clear outline of the study's objectives and assured that their responses would be kept confidential. The questionnaire was distributed via an online survey platform designed to facilitate non-probabilistic sampling for remote data collection. To maintain sample consistency, eligibility was restricted to individuals aged 18 and older residing in Peru. The selection of Peru as a study context is based on a combination of methodological feasibility and theoretical relevance. From a practical perspective, it ensured efficient access to the target population and a deep understanding of the local banking sector context, critical factors for the instrument's design and validation. Academically, Peru represents a strategic case study as an emerging economy with rapid adoption of digital banking but moderate overall institutional trust. This context allows examination of trust-building mechanisms in automated services in an environment where technological factors and uncertainty are pronounced, offering insights that may differ from findings in economies with more mature digital markets or higher baseline trust. Therefore, this approach meets feasibility criteria and enables exploration of theoretical relationships in a scenario where uncertainty reduction and digital trust-building are paramount. The collection was carried out by inviting people online. The initial collection phase yielded 407 responses. Following this, a rigorous data-cleaning process was undertaken, which involved verifying each participant's consent status, confirming their residential location, and carefully reviewing the logical

consistency of their answers. Responses that demonstrated attentiveness and coherence were retained, while those lacking these qualities were excluded. Ultimately, this stringent selection process refined the dataset to 362 fully qualified responses, forming the final sample for subsequent data analysis.

4.4. Ethical statement

This research did not include any variable manipulations that could have affected participants' physical or mental health. No collection of sensitive or personal data occurred that might have compromised their privacy or security. Instead, the study was strictly observational, focused on recording behaviors as they naturally unfolded in a public environment, with individuals acting independently and without interaction with researchers. The observations were limited to general behavioral trends and did not involve any data that could lead to personal identification. The scope was confined to a publicly accessible area, ensuring that participant safety, privacy, and overall well-being remained protected. As no procedures were implemented that might influence participants' health, safety, or autonomy, there was no ethical requirement to seek review or approval from an ethics committee.

4.5. Analysis of data

Partial Least Squares Structural Equation Modeling (PLS-SEM) through SmartPLS software version 4.1.0.2. To evaluate the internal consistency of each construct, Cronbach's Alpha (CA) and composite reliability (CR) were calculated, where a minimum threshold of 0.7 is considered adequate for reliable measurement. The Average Variance Extracted (AVE) was also assessed to determine the degree to which a construct captures variance relative to the variance attributed to measurement error. An AVE value of at least 0.5 is desirable, indicating that the construct accounts for at least half of the variance of its indicators [47]. To ensure discriminant validity, the Fornell-Larcker criterion was applied. This criterion requires that the square root of the AVE for each construct exceeds any correlations with other constructs, confirming that each construct is uniquely represented within the model [48]. The Heterotrait-Monotrait Ratio (HTMT) was calculated with an acceptable value below 0.9, indicating adequate discriminant validity, signifying that constructs are sufficiently distinct from one another [47]. The Variance Inflation Factor (VIF) assessment is a key tool for detecting common method bias (CMB) in PLS-SEM. This analysis is performed by calculating the VIF values in the internal model using a full collinearity test. When all values obtained are less than or equal to 5, the model is considered to have no evidence of CMB, allowing us to proceed with greater confidence in the results [49]. To test the hypotheses, both direct and indirect effects were computed, and statistical significance was assessed using bootstrapping with 10,000 resamples. This approach provided a robust examination of the relationships among variables within the proposed model.

5. Results

This section presents the study's results, beginning with a demographic analysis of the participants that provides context and reflects the sample's diversity. This is followed by a description of reliability test results to ensure the internal consistency of the measures, as well as analyses of convergent and discriminant validity, which corroborate the accuracy of the variables used in the model. Finally, the direct and indirect (mediating) effects of the proposed model are examined, providing insight into the relationships between the variables and their influence on the results.

5.1. Demographic information

The study was carried out with a sample of 362 participants. The

participants' ages ranged from 18 to 65 years, with an average of 27.31 years. Of the participants, 196 (53.99 %) were women. Regarding education, 207 (57.02 %) were higher education students, 68 (18.73 %) were higher education completers, 60 (16.52 %) had postgraduate completers, and 7 (1.92 %) corresponded to other categories of studies. In relation to monthly income, 163 (44.90 %) had incomes lower than the Peruvian Minimum Wage (PMW) (1025 soles; 275 USD), 74 (20.38 %) had incomes between 1 and 2 PMW, 34 (9.36 %) had incomes between 2 and 3 PMW, 39 (10.74 %) earned between 3 and 4 PMW, and 52 (14.32 %) received >4 PMW. Table 1 presents the results obtained in terms of the reliability and validity of the variables assessed.

5.2. Reliability and validity

The coefficients obtained meet the minimum values expected for CA, CR, and AVE. Most items have loadings above 0.5, indicating they adequately represent each variable's dimensions. These results indicate that the selected items are adequate and contribute significantly to explaining the constructs included in the model. VIF is also shown in Table 1.

Composite reliability rho_a for each variable is presented in Fig. 1. It is shown that the values were above 0.7.

Fig. 2 presents the values of composite reliability rho_c for each variable. The values were >0.7.

The Average Variance Extracted (AVE) of the variables in the research model is presented in Fig. 3.

5.3. Common method bias

Common method bias occurs when the results of a research study are influenced by the way the data are collected, rather than accurately reflecting the true relationship between the concepts being studied. When a single method is used to measure several variables (e.g., a survey in which participants respond to different aspects of the same questionnaire), artificial correlations may appear between these variables. These correlations do not reflect real relationships between the concepts but rather the effects of the measurement instrument itself. According to Kock [49], the common bias method is evaluated using collinearity statistics (VIF) in the inner model, expecting values of five or less. In Table 2, all values are <5, indicating the absence of common method bias.

5.4. Discriminant validity

It is a property of a measurement model that indicates how well a construct is distinct from other related constructs. Thus, it verifies that one construct does not overlap with another, ensuring that each latent variable measures something unique within the model. The Fornell-Larcker criterion [48] is employed to confirm discriminant validity, while the HTMT [47] is also used to verify it. Table 3 presents the verification of discriminant validity using the Fornell-Larcker criterion, while Table 4 provides an evaluation by the HTMT. In both cases, discriminant validity is verified.

5.5. R^2 and Q^2 values

The R^2 measures how well an independent variable explains the variability of a dependent variable in a regression model. Its value ranges from 0 to 1. When it is closer to 1, it implies that the model explains the variability of the dependent variable well; conversely, a value closer to 0 indicates that the model does not explain the variability well. According to Sarstedt et al. [50], values of R^2 equal to or greater than 0.5 indicate a moderate relationship. The Q^2 metric evaluates the model's predictive ability using cross-validation. If Q^2 is more than zero, it means that the model has predictive relevance. Higher values indicate better predictive ability. The R^2 values were greater than 0.5, indicating a

Table 1
Reliability and validity.

Variables	Factor loading	VIF	Reliability and validity
Information quality			
My bank's chatbot always provides me with the information I need	0.819	2.434	CA: 0.915
My bank's chatbot always answers my questions as expected	0.871	3.121	CR (rho_a): 0.916
My bank's chatbot always provides me with sufficient information	0.848	3.044	CR (rho_c): 0.934
The information from my bank's chatbot is always useful to solve my problems	0.866	3.001	AVE: 0.703
The information from my bank's chatbot is always reliable	0.798	2.280	
The information from my bank's chatbot is always easy to understand	0.825	2.469	
System quality			
I found it easy to become proficient in using my bank's chatbot.	0.831	2.734	CA: 0.873
I find my bank's chatbot easy to use	0.858	2.975	CR (rho_a): 0.886
Using my bank's chatbot required minimal mental effort	0.627	1.421	CR (rho_c): 0.905
My bank's chatbot responded quickly	0.808	2.114	AVE: 0.615
My bank's chatbot is always available	0.797	2.049	
I am satisfied with the chatbot	0.762	1.658	
Satisfaction			
The chatbot has done a good job	0.930	3.433	CA: 0.926
The chatbot did what I expected	0.929	3.543	CR (rho_a): 0.928
I am happy with the chatbot	0.942	3.865	CR (rho_c): 0.953
Trust			
My bank's chatbot is reliable	0.847	2.238	CA: 0.909
My bank's chatbot is competent and intelligent	0.882	2.589	CR (rho_a): 0.911
My bank's chatbot is pleasant	0.909	3.937	CR (rho_c): 0.937
My bank's chatbot is friendly	0.910	3.929	AVE: 0.787
Loyalty			
I am a loyal user of my bank's chatbot	0.888	2.878	CA: 0.920
I intend to continue to be a user of my bank's chatbot.	0.919	3.702	CR (rho_a): 0.926
I will continue to use my bank's chatbot even though I can use my phone banking	0.906	3.978	CR (rho_c): 0.943
I will continue to use my bank's chatbot even though I may use my bank's app	0.880	3.272	AVE: 0.807
e-WOM			
I have recommended my bank's chatbot to several people.	0.935	3.515	CA: 0.910
I put in a good word for my bank's chatbot to my friends	0.929	3.238	CR (rho_a): 0.918
I advertise my bank's chatbot a lot	0.897	2.688	CR (rho_c): 0.943
			AVE: 0.848

moderate relationship. Additionally, using PLSpredict, the Q^2 values were calculated and were positive, indicating good predictive relevance (Table 5).

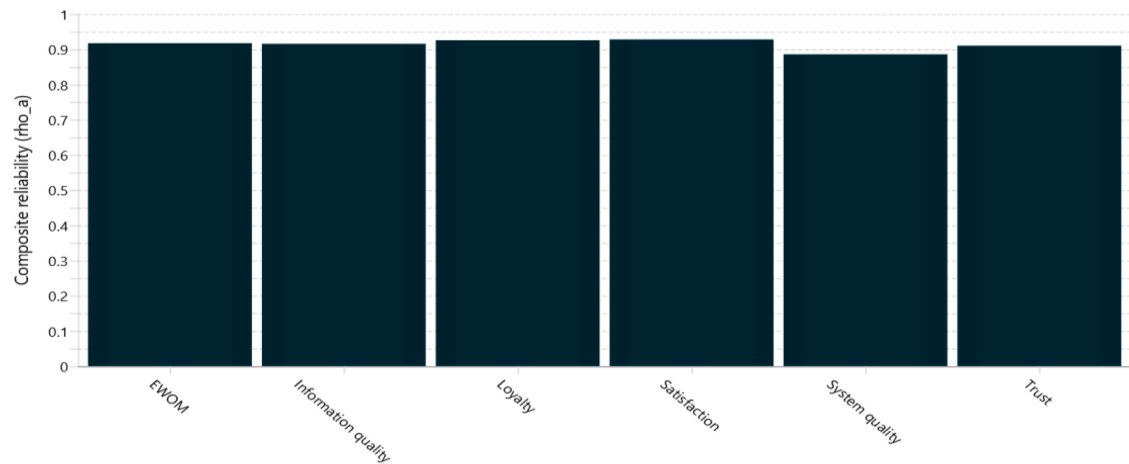


Fig. 1. Composite reliability rho_a per construct.

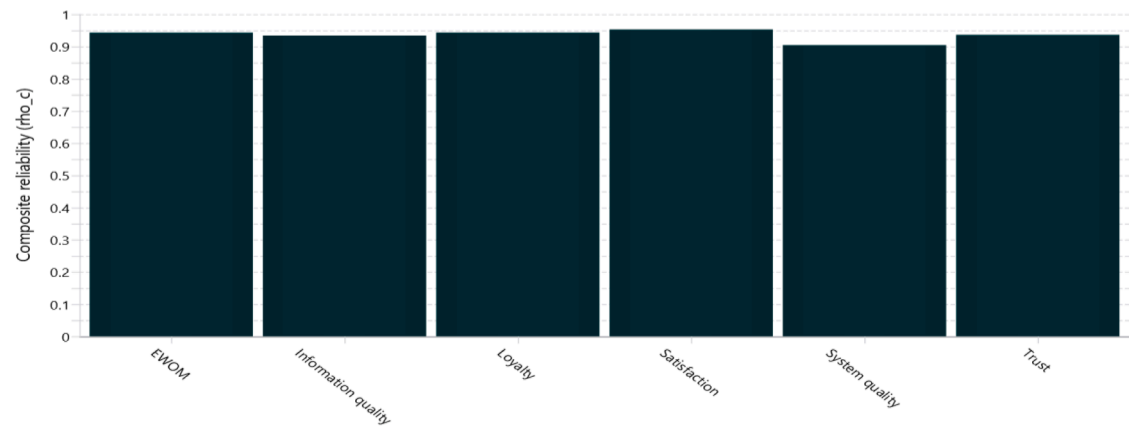


Fig. 2. Composite reliability rho_c per construct.

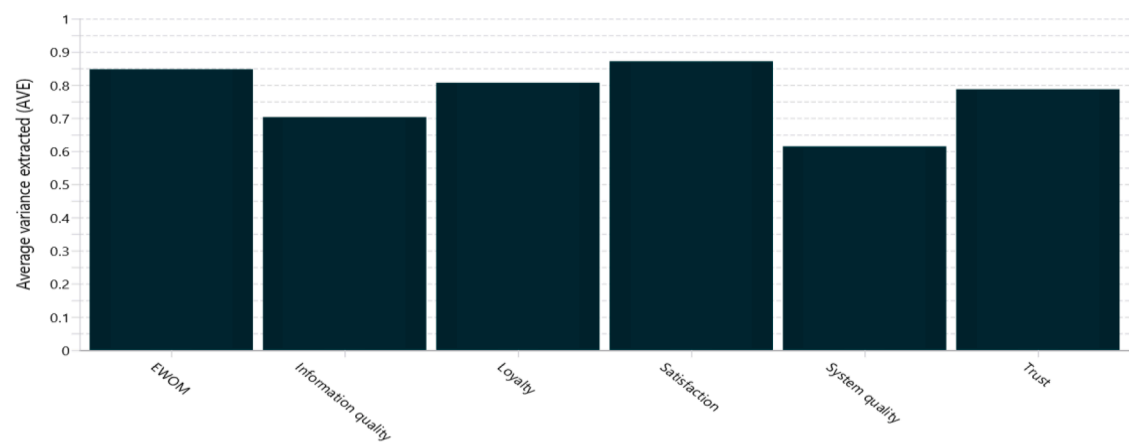


Fig. 3. Average Variance Extracted (AVE) per construct.

5.6. Bootstrapping

It is a statistical resampling technique used to assess the significance of the effects. After 10,000 iterations, all p-values were <0.05, indicating that the observed effects between variables are statistically significant (i.e., they did not occur by chance; Table 6). The mediation effect of a model is the ability to explain the mechanism by which an independent variable (X) influences a dependent

variable (Y) via a third, intermediate variable, known as the mediating variable (M). Table 7 presents the model's mediation effect, which enables measurement of specific indirect effects. The p-values were below 0.05, indicating statistical significance. Fig. 4 shows the values of the effects on the serial mediation of trust and satisfaction, included in the research model. The results of the tested research model are shown in Fig. 5.

Table 2

Collinearity statistics (VIF) - Inner model.

	e-WOM	Information quality	Loyalty	Satisfaction	System quality	Trust
e-WOM						
Information quality						2.771
Loyalty						
Satisfaction	1.000		1.000			
System quality						2.771
Trust				1.000		

Table 3

Discriminant validity - Fornell-Larcker criterion.

	e-WOM	Information quality	Loyalty	Satisfaction	System quality	Trust
e-WOM	0.921					
Information quality	0.714	0.838				
Loyalty	0.759	0.723	0.898			
Satisfaction	0.711	0.828	0.728	0.934		
System quality	0.553	0.793	0.647	0.773	0.784	
Trust	0.595	0.793	0.675	0.812	0.821	0.887

Table 4

Discriminant validity - Heterotrait-monotrait ratio (HTMT).

	e-WOM	Information quality	Loyalty	Satisfaction	System quality	Trust
e-WOM						
Information quality	0.723					
Loyalty	0.766	0.721				
Satisfaction	0.702	0.879	0.717			
System quality	0.543	0.866	0.656	0.858		
Trust	0.569	0.824	0.664	0.852	0.895	

Table 5R² and Q² values.

Variables	R ²	R ² adjusted	Q ²
e-WOM	0.505	0.504	0.400
Loyalty	0.530	0.528	0.465
Satisfaction	0.659	0.658	0.682
Trust	0.728	0.727	0.723

6. Discussion

Table 8 presents a comparison of outcomes from the current research and previous studies.

The results confirm that the effects of information and system quality on loyalty and e-WOM are not direct but rather mediated sequentially by trust and satisfaction. This finding supports the psychological logic underlying our integrated model: perceived quality initially reduces uncertainty and enables positive attributions, based on URT and AT, thereby generating trust, which establishes expectations that, when confirmed (EDT), produce satisfaction as the overall evaluation. The

Table 6

Direct effects of the variables.

	Original sample	Original mean	Standard deviation	T statistics	p-values	f ²	Decision
H1: Information quality → Trust	0.382	0.381	0.050	7.557	0.000	0.199	Accept
H2: System quality → Trust	0.519	0.519	0.051	10.260	0.000	0.367	Accept
H3: Trust → Satisfaction	0.812	0.812	0.020	39.750	0.000	1.933	Accept
H4: Satisfaction → Loyalty	0.728	0.728	0.038	19.124	0.000	1.126	Accept
H5: Satisfaction → e-WOM	0.711	0.711	0.025	28.306	0.000	1.021	Accept

Table 7

Specific indirect effects of the variables.

	Original sample	Original mean	Standard deviation	T Statistics	p-values	Decision
H6: IQ → Trust → Satisfaction	0.310	0.309	0.043	7.269	0.000	Accept
H7: SQ → Trust → Satisfaction	0.421	0.422	0.043	9.796	0.000	Accept
H8: IQ → Trust → Satisfaction → Loyalty	0.225	0.225	0.031	7.164	0.000	Accept
H9: IQ → Trust → Satisfaction → e-WOM	0.220	0.220	0.033	6.730	0.000	Accept
H10: SQ → Trust → Satisfaction → Loyalty	0.306	0.307	0.040	7.751	0.000	Accept
H11: SQ → Trust → Satisfaction → e-WOM	0.299	0.300	0.033	9.170	0.000	Accept
H12: Trust → Satisfaction → Loyalty	0.591	0.591	0.037	9.170	0.000	Accept
H13: Trust → Satisfaction → e-WOM	0.577	0.578	0.028	20.336	0.000	Accept

IQ: Information quality; SQ: System quality.

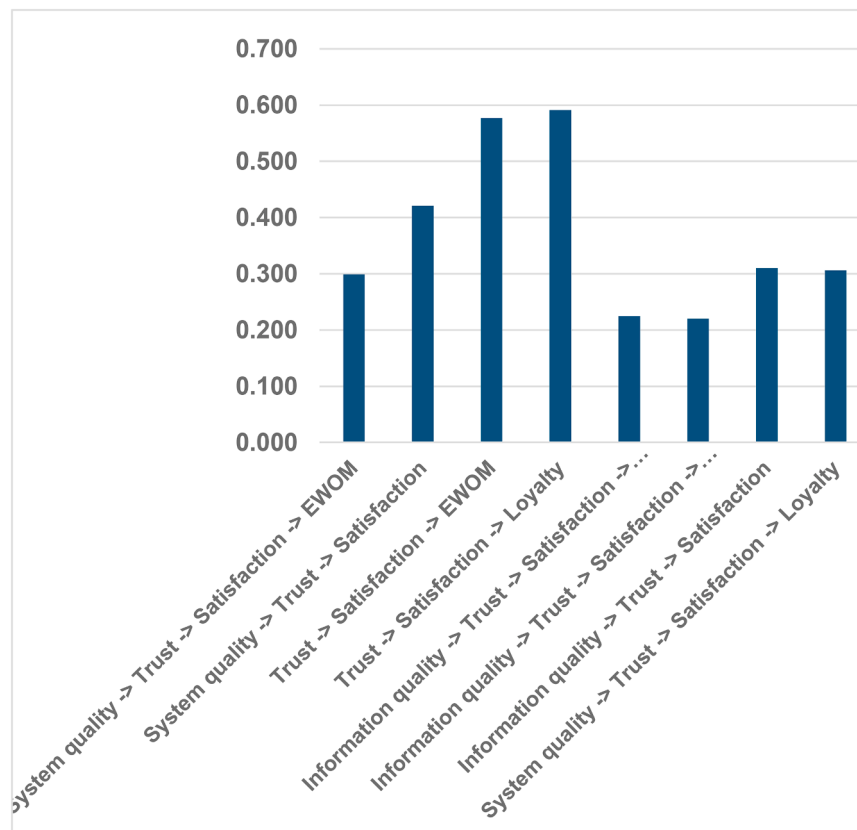


Fig. 4. Serial mediation effects of trust and satisfaction.

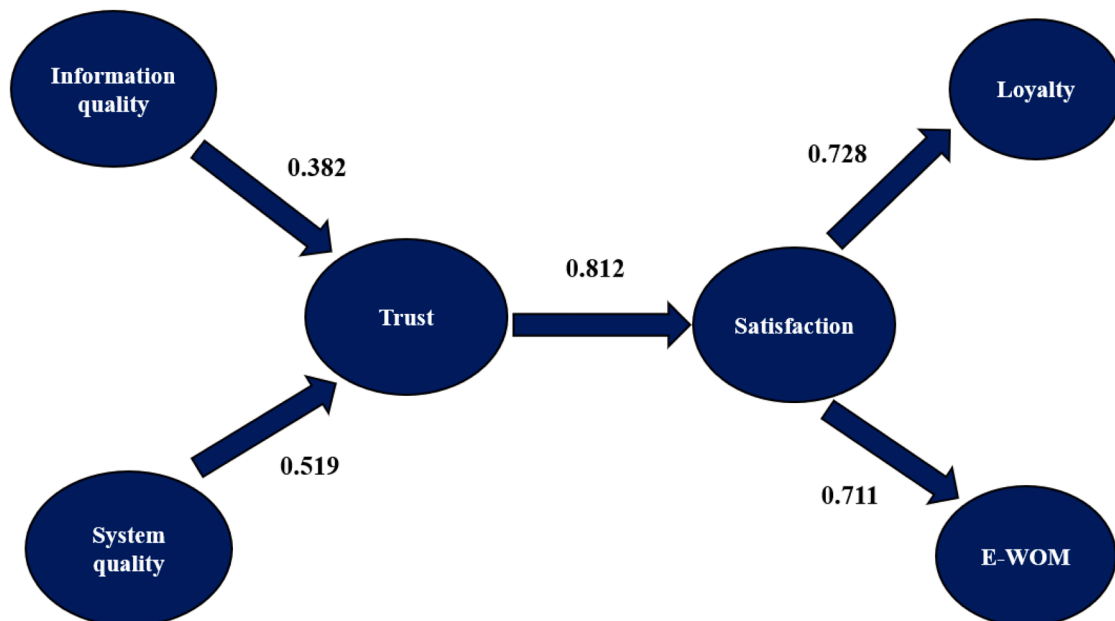


Fig. 5. Tested Structural Model (PLS-SEM).

inclusion of both mediators thus captures the entire process from technical perception to behavioral engagement, explaining how quality translates into relational outcomes through intermediate psychological states. This mediating cadence is consistent with recent findings in chatbot research and refines our understanding of the mechanisms linking technical expertise to customer loyalty.

6.1. Theoretical implications

This study provides clear theoretical implications by integrating and expanding four fundamental theoretical frameworks for explaining user experience with chatbots, addressing the gap in how traditional socio-cognitive mechanisms are reconfigured into an integrated model. URT is extended by demonstrating that, in human-chatbot interactions,

Table 8

Comparison of outcomes between the current and previous study.

	Beta	p-values	Sources
H1: Information quality → Trust	0.037	$p > 0.05$	Chandra and Tan [12]
	0.193	$p < 0.001$	Mbawuni and Nimako [51]
	0.236	$p < 0.05$	Manan et al. [19]
	0.312	$p < 0.05$	Nelloh et al. [9]
	0.382	$p < 0.001$	Current study
H2: System quality → Trust	0.432	$p < 0.001$	Setiyono et al. [10]
	0.081	$p > 0.05$	Manan et al. (2018)
	0.255	$p < 0.05$	Chandra and Tan [12]
	0.369	$p < 0.01$	Ludwina et al. [11]
	0.519	$p < 0.001$	Current study
H3: Trust → Satisfaction	0.248	$p < 0.001$	Omorieg et al. [16]
	0.278	$p < 0.001$	Abdennebi [52]
	0.300	$p < 0.001$	Leclercq-Machado et al. [13]
	0.360	$p < 0.05$	Susanti and Azis [53]
	0.367	$p < 0.001$	Murdefin et al. [54]
H4: Satisfaction → Loyalty	0.429	$p < 0.001$	Mondego and Gide [14]
	0.443	$p < 0.001$	Damberg [55]
	0.812	$p < 0.001$	Current study
	0.224	$p < 0.01$	Ahmed et al. [56]
	0.370	$p < 0.01$	Fauzi and Suryani [57]
	0.525	$p < 0.01$	Albaity and Rahman [15]
	0.591	$p < 0.001$	Haron et al. [58]
	0.628	$p < 0.001$	Omorieg et al. [16]
	0.675	$p < 0.001$	Leclercq-Machado et al. [13]
	0.728	$p < 0.001$	Current study
H5: Satisfaction → e-WOM	0.247	$p < 0.001$	Salah et al. [18]
	0.466	$p < 0.001$	Giao et al. [59]
	0.596	$p < 0.01$	Ugwuanyi et al. [60]
	0.705	$p < 0.01$	Luu Thi Thuy et al. [17]
	0.711	$p < 0.001$	Current study

technological attributes, information quality, and system quality replace interpersonal cues as primary mechanisms for reducing uncertainty. The findings reveal that trust is constructed asymmetrically: the user infers trustworthiness solely from technical performance and informational accuracy, shifting the focus of URT from interpersonal competence to technological credibility. This suggests that URT models in technology contexts should be recalibrated to incorporate system transparency and explainability as relevant dimensions. EDT is enriched by examining the formation of satisfaction in non-human interactions, which allowed for the understanding that, in the absence of human presence, trust shapes the recalibration of expectations. The strong relationship between trust and satisfaction indicates that, in chatbot-based services, the evaluation of the experience is driven more by system efficiency and information credibility than by traditional social comparisons. This expands the EDT by highlighting the mediating role of trust in evaluating expectations in automated environments.

We refine the SDT by showing that, while chatbots satisfy the needs for autonomy and competence, their limited capacity to establish emotional connections restricts the development of deeper social relationships. This suggests that SDT in technology contexts must consider the asymmetry in satisfaction, where technical competence can partially compensate for, but not replace, the need for human connection. TA is applied to explain inconsistencies in the previous literature. Our results indicate that users attribute service quality to internal or external factors, which explains the formation of trust and satisfaction. This provides an explanatory framework for contradictory findings in previous studies and highlights the importance of managing causal attributions in chatbot design. All these contributions narrow the gap identified in the literature, providing greater insight into the lack of an integrated framework for explaining the sequential psychological mechanisms underlying chatbot interactions. By demonstrating how URT, EDT, SDT, and TA complement each other to explain the stages of uncertainty reduction, expectation management, motivation, and causal attribution, we offer a more comprehensive theoretical model for understanding

how users evaluate and respond to conversational systems such as chatbots. This integrated model lays the groundwork for future research exploring other variables.

6.2. Managerial implications

This study offers key implications for optimizing chatbots in customer service. Given that information quality ($\beta = 0.382$) and system quality ($\beta = 0.519$) significantly impact trust, companies must invest in AI to provide accurate and personalized responses, while ensuring speed and stability to avoid user frustration. Trust, a key mediator of satisfaction ($\beta = 0.812$), should be strengthened through transparency, explaining the origin of chatbot responses, and prioritizing data security with encryption protocols. A hybrid approach that routes complex queries to human agents also strengthens trust. The positive impact of satisfaction on loyalty ($\beta = 0.728$) and digital word-of-mouth (e-WOM, $\beta = 0.711$) suggests focusing on exceptional experiences. This is achieved by analyzing user behavior with machine learning to optimize interactions and by implementing gamification strategies or loyalty programs.

The confirmation that satisfaction and trust mediate the relationship between chatbot quality and loyalty ($\beta = 0.225$; $\beta = 0.306$) and e-WOM ($\beta = 0.220$; $\beta = 0.299$) implies that design strategies must continually consider how interactions generate these perceptions. Continuous measurement and adjustments are essential. Furthermore, it is crucial to address ethical implications. Satisfaction mediates the relationship between trust and loyalty ($\beta = 0.591$), so chatbots must provide truthful and reliable information. Likewise, by mediating between trust and e-WOM ($\beta = 0.577$), any failure in trustworthiness can generate negative comments and damage reputation.

6.3. Limitations and future studies

The main limitation is its geographic and sectoral focus, as it is restricted to the Peruvian banking sector. This specificity limits the generalizability of the findings, as cultural, regulatory, and consumer expectations can vary significantly across countries and industries, potentially introducing sample bias. Furthermore, the study presents methodological limitations that warrant caution in interpreting the findings. Cross-sectional design prevents making strong causal inferences and capturing the temporal dynamics of the proposed relationships, such as the evolution of trust over repeated interactions. The reliance on self-reported data carries the risk of common-method bias and social desirability bias. Furthermore, the sample, while suitable for PLS-SEM, may be subject to self-selection bias, with users who are more technologically adept or have more positive prior experiences more likely to participate. These factors limit the generalizability of the results beyond the Peruvian banking context and users with a certain level of digital adoption. Therefore, future research should adopt longitudinal or experimental designs to strengthen causal validity and incorporate objective measures of usage or behavior to complement perceptions. Replicating the model across diverse sectors, such as retail, healthcare, telecommunications, and cultural contexts, is also recommended. This would allow for the examination of effects in these other scenarios, thus enriching the theoretical and practical framework.

7. Conclusions

The study confirms that the quality of the information and system in chatbots is fundamental to generating trust and satisfaction in users, which positively impacts loyalty and e-WOM. The research shows that trust and satisfaction not only reinforce customer loyalty to the brand but also promote digital recommendations, strengthening the company's image and positioning in a highly competitive market. Additionally, Uncertainty Reduction Theory has significant applications in digital environments. Interaction with high-quality chatbots enables the

reduction of user uncertainty, thereby reinforcing trust in technology. This finding extends the theory to the context of artificial intelligence, where system quality and information are key tools to facilitate effective and predictable interactions. Similarly, the Expectation Confirmation Theory is validated in the digital domain, showing that meeting or exceeding users' expectations in their interactions with chatbots increases satisfaction and the likelihood of positive e-WOM. Further empirical analysis reveals that satisfaction derived from using chatbots is significantly associated with users' loyalty and willingness to recommend the service. This mediating effect on satisfaction and trust underlines the crucial role of quality in digital services in establishing long-term customer relationships. This study contributes to the digital marketing literature by providing an expanded view of the factors that influence customer experience on digital platforms. It opens up new research opportunities on the impact of artificial intelligence on customer satisfaction and loyalty.

Declaration of generative AI and AI-assisted technologies in the writing process

Nothing to declare

Appendix 1

N°		1	2	3	4	5
Information quality						
1	My bank's chatbot always provides me with the information I need					
2	My bank's chatbot always answers my questions as expected					
3	My bank's chatbot always provides me with sufficient information					
4	The information from my bank's chatbot is always useful to solve my problems					
5	The information from my bank's chatbot is always reliable					
6	The information from my bank's chatbot is always easy to understand					
System quality						
7	I found it easy to become proficient in using my bank's chatbot.					
8	I find my bank's chatbot easy to use					
9	Using my bank's chatbot required minimal mental effort					
10	My bank's chatbot responded quickly					
11	My bank's chatbot is always available					
12	I am satisfied with the chatbot					
Satisfaction						
13	The chatbot has done a good job					
14	The chatbot did what I expected					
15	I am happy with the chatbot					
Trust						
16	My bank's chatbot is reliable					
17	My bank's chatbot is competent and intelligent					
18	My bank's chatbot is pleasant					
19	My bank's chatbot is friendly					
Loyalty						
20	I am a loyal user of my bank's chatbot					
21	I intend to continue to be a user of my bank's chatbot.					
22	I will continue to use my bank's chatbot even though I can use my phone banking					
23	I will continue to use my bank's chatbot even though I may use my bank's app					
e-WOM						
24	I have recommended my bank's chatbot to several people.					
25	I put in a good word for my bank's chatbot to my friends					
26	I advertise my bank's chatbot a lot					

Data availability

Data will be made available on request.

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CRediT authorship contribution statement

Diego Cardenas-Pauckar: Writing – review & editing, Writing – original draft, Visualization, Resources, Project administration, Methodology, Investigation. **Shyla Del-Aguila-Arcenales:** Writing – review & editing, Writing – original draft, Resources, Project administration, Methodology, Investigation. **Enzo Puente-Custodio:** Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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